Maciej Krzysztof Zuziak

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LinkedIn: maciej-zuziak | GitHub: MKZuziak | Scholar: Google Scholar

Experience

- **Institute of Science Tokyo** — *Oct 2026 – Dec 2026*
  Visiting Researcher (Incoming), School of Computing — Tokyo, Japan
- **European University Institute (EUI)** — *Sep 2026 – Sep 2027*
  Max Weber Fellow, Department of Law — Florence, Italy
- **University of Leeds** — *Jan 2026 – Jun 2026*
  Research Fellow, School of Computer Science — Leeds, United Kingdom
- **National Research Council of Italy (CNR)** — *Nov 2021 – Dec 2025*
  Research Associate, Knowledge Discovery Lab — Pisa, Italy
- **Tellu AS** — *Sep 2023 – Sep 2024 (dual appointment)*
  Visiting Research Engineer, DevOps Department — Oslo, Norway

Education

- **University of Pisa** — *Nov 2021 – Dec 2025*
  Department of Computer Science, Ph.D. (Artificial Intelligence) — Pisa, Italy
- **University of Silesia in Katowice** — *Sep 2016 – Jul 2021*
  Faculty of Law and Administration, Magister Prawa (Master of Laws) — Katowice, Poland

Journals

First = first author; Co-first = equal contribution, as marked in the publication

[1] [First] **M.K. Zuziak**, R. Pellungrini, S. Rinzivillo (2026): **One-Shot Clustering for Federated Learning Under Clustering-Agnostic Assumption**. ACM Transactions on Intelligent Systems and Technology (accepted subject to minor revisions).

Conference Papers

First = first author; Co-first = equal contribution, as marked in the publication

[1] [Co-first] S.Z. El Mestari*, **M.K. Zuziak***, G. Lenzini (2026): **Poison to Detect: Detection of Targeted Overfitting in Federated Learning**. *Proceedings on Privacy Enhancing Technologies* 2026(4), pp. 480–506. Presented at *PETS 2026*, Calgary, Canada. (*equal contribution)

[2] J. Nikolic, **M.K. Zuziak**, E. Pournaras (2026): **The Optimization Trilemma: Efficiency, Comfort and Fairness in Decentralized Multi-agent Coordination**. At *IEEE ACSOS 2026*, Cesena, Italy. IEEE (to appear in IEEE Xplore).

[3] [Co-first] S.Z. El Mestari*, **M.K. Zuziak***, G. Lenzini, S. Rinzivillo (2025): **Can Contributing More Put You at a Higher Leakage Risk? The Relationship Between Shapley Value and Training Data Leakage Risks in Federated Learning**. At *SECRYPT 2025*, Bilbao, Spain. SciTePress. (*equal contribution)

[4] [First] **M.K. Zuziak**, R. Pellungrini, S. Rinzivillo (2024): **One-Shot Clustering for Federated Learning**. At *IEEE Big Data 2024*, Washington, USA. IEEE Xplore.

[5] [First] **M.K. Zuziak**, S. Rinzivillo (2024): **Amplified Contribution Analysis for Federated Learning**. At *IDA 2024*, Stockholm, Sweden. Springer Lecture Notes in Computer Science (LNCS).

[6] [Co-first] **M.K. Zuziak***, O. Hinrichs*, A. Abdrassulova, S. Rinzivillo (2023): **Data Collaboratives with the Use of Decentralised Learning**. At *ACM FAccT 2023*, Chicago, USA. ACM. (*equal contribution)

Workshops & Posters

First = first author; Co-first = equal contribution | W = Workshop paper; P = Poster

[W.2] [First] **M.K. Zuziak**, S.Z. El Mestari (2026): **Personal Data, Generated**. Accepted poster at *NeurIPS 2026 Workshop on Privacy in the Era of Large Opaque Models*, Paris, France (non-archival).

[P.3] [Co-first] S.Z. El Mestari*, **M.K. Zuziak***, G. Lenzini (2026): **Poison to Detect: Detection of Targeted Overfitting in Federated Learning (Poster Version)**. Privacy Enhancing Technologies Symposium, Canada. (*equal contribution)

Awards

Max Weber Fellowship, EUI (2026, under 3%) · Frontier AI Security Training (FAST), SASH (2026, under 3%) · Horizon Europe (Project Tango) Fellowship, CNR (2024) · Marie Skłodowska-Curie Actions Fellowship (2021, under 10%) · CSC Scholarship (2021) · Academic Scholarship, University of Silesia (top 5%, 2017)

Languages

Polish (Native), English (C2), Chinese (B2), Italian (B1/B2), Japanese (A2)

Skills

Core: Federated Learning, Privacy-Preserving ML, Explainable AI, Data Governance. **Programming:** Python, Julia, C/C++, TypeScript. **ML & Data:** PyTorch, TensorFlow/Keras, JAX, Scikit-Learn, NumPy, Pandas. **Systems:** Flower, PySyft/PyGrid, TensorFlow Federated, Docker, Kubernetes, Linux/WSL. **HPC:** EuroHPC (MeluXina, Leonardo Booster). **LLM Systems:** LangChain, LangGraph, RAG, vector databases (FAISS, Pinecone, Weaviate), LoRA/PEFT.